

Dept. of Computer and Communication Engineering
Patuakhali Science and Technology University
6th Semester (Level-3, Semester-II), Midterm Examination of B.Sc. Engg. (CSE), July-December: 2021
Course Code: CCE-321 Course Title: Computer Peripheral and Interfacing
Credit Hour: 3.0 Full Marks: 15 Duration: 60 Minutes

1. ~~a)~~ Define software interrupt with example.
- ~~b)~~ What are the problems of linear decoding system in main memory array design? How to solve them?
- ~~c)~~ Interface two chips of 4k RAM and 1 chip of 2k ROM with processor. Starting address of RAM is 8000H.
2. ~~a)~~ What is Usability in HCI?
- ~~b)~~ Why is usability important in HCI? Give an example.
- ~~c)~~ Write down the Norman's Seven Principles for HCI design.

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Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Peripherals

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2018-2019

Course Code
CCE-321

Course Title
Computer Peripherals and Interfacing

July-December 2021

Credit: 03

Time: 03 Hr

Marks: 70

- Answer any 05 out of 06 Questions (Split answers are highly discouraged)
1. [A.] What are the different components of IoT? How does the Internet of Things (IoT) affect our everyday lives? 4
 - [B.] What is communication model in IoT? Explain different types of communication model in IoT with diagram and example. 6
 - [C.] Write down the difference between Embedded and Real Time Systems. 4
 2. [A.] What is the periodic and aperiodic task model in real-time system? Explain with example. 4
 - [B.] What is a RVDT? How does RVDT work step by step? Explain with RVDT construction diagram. 4
 - [C.] How many principles are there in Norman design principles? Write down Norman's design principles for IoT. 3
 - [D.] Briefly explain the most 6 communication protocols used by IoT. 3
 3. [A.] Explain the characteristics of RTS. 3
 - [B.] Write down the difference between Sensors and Transducer. 3
 - [C.] What is Human Computer Interaction? Briefly explain most important components of HCI. 5
 - [D.] What are the types of IoT security? How to Secure IoT Devices? 3
 4. [A.] a. What do you mean by true color representation? Does it represent any disadvantages? How to handle the problems? 5
 - b. If we use 5 bits each for red and blue and 6 bits for green in direct coding method, how many possible simultaneous colors do we have? 5
 5. Interpret EFE_{16} as unsigned, signed, and floating point numbers that a microprocessor can understand. 4
 6. What are the common problems with scanner? Share some tips to troubleshoot the basic scanner related problems. 3
 - Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output port and port B as input port and port C as output port. 2
 5. [A.] How do the USB peripherals communicate with microprocessor? Give the pin-out for USB type C 3.0. 5
 - [B.] Categorize and explain the operational modes of 8279. How many ways the keyboard is interfaced with the CPU? 4
 - [C.] The-microcomputer communicates with an external device via one or more registers called I/O ports using programmed I/O. Illustrate the functions of two of them with brief description. 3
 - [D.] Make a comparison between buffers and spoolers. 2
 6. [A.] Suppose, a processor is having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor. 5
 - a. What is the memory capacity of the processor?
 - b. How many address lines are required for a RAM?
 - c. How many RAM chips are required?
 - d. What will be the size of address decoder?
 - [B.] Show the software keyboard interfacing process through a flowchart. Describe each step for a 4x4 keyboard. 4
 - [C.] Serialize the steps associated in turning on or off a pixel in LCD monitor. 3
 - [D.] An interface incorporates several components including its communication type as well as the interface software. List out eight major communication types in case of a printer. 2

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Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: II Session: 2017-2018
Course Code CCE-321 Course Title Computer Peripheral and Interfacing July-December 2020 Credit: 03
Time: 03 Hr Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

- 1 [A.] Enlist inside parts of optical mouse. How does an optical computer mouse work? 5
- [B.] What is Impact Printers? How does Impact Printers work? Explain with functional diagram. 6
- [C.] Draw the functional block diagram of 8251A PCI. 3
- [A.] What determines the quality of monitor? 3
- [B.] What is a RVDT? How does RVDT work step by step? Explain with RVDT construction diagram. 6
- [C.] What is RS-232 Handshaking? Explain with figure. 2
- [D.] Differentiate between span and range of a transducer system. 3
- [A.] a) What is a plasma monitor? 3
- b) What is nonlinearity error? 3
- [B.] Write down the difference between Sensors and Transforms. 3
- [C.] How does a process control loop work? Briefly explain with figure. 4
- [D.] Assume the ASCII character "A" (01000001), is framed in between the start bit and two stop bits. Notice that the LSB is sent out first. Now answer the following questions. 4
 - i. Define character-oriented transmission.
 - ii. How the data is packed?
 - iii. How many bits constitute the character?
 - iv. When the data begins and ends?
 - v. Distinguishes between Synchronous and Asynchronous Serial Communication.
 - vi. Which communication method is suitable for this transmission?
 - vii. Explain the character transmission with framing diagram.
- [A.] Show the pin configuration of 8255 PPI and describe the operation performed by each unit separately. 6
- [B.] Convert 39887.5625 to IEEE 32-bit floating point format. 4
- [C.] Explain software keyboard interfacing procedure with circuit connections and algorithm. 4
- [A.] Draw the architectural diagram of 8279 programmable keyboard/display controller. How does 8279 keyboard work? 6
- [B.] Differentiate between programmed I/O and interrupt initiated I/O. 4
- [C.] How can the signed binary numbers be represented? 3
- [D.] What is the range of signed values that may be stored in 20 bits? 1
- [A.] How to connect a microprocessor to ROM/RAM chips? Illustrate and explain each technique. 9
- [B.] When is an overflow flag set or reset? Give examples. 3
- [C.] What are the basic principles of two-wire handshaking method of data transfer? 2

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Peripheral

Patuakhali Science and Technology University
Final Examination of B.Sc. Engg. (CSE), Level: 3 Semester: II
Course Code: CCE 321 Course Title: Computer Peripheral and Interfacing
Credit Hour: 3.00 Full Marks: 70 Duration: 03 Hours

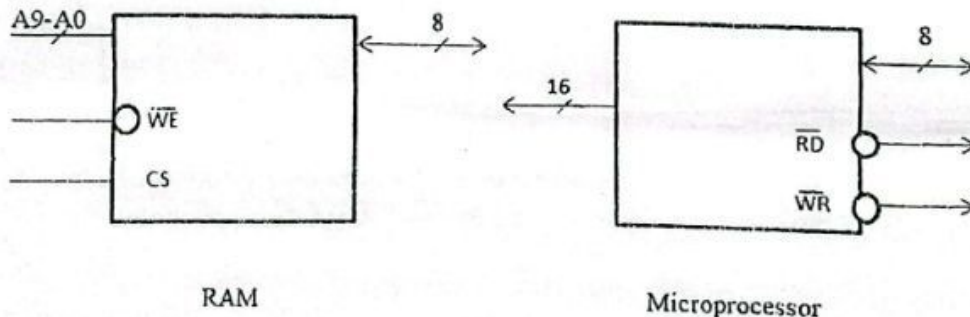
[Figures in the right margin indicate full marks. **Split answering of any question is not recommended.** Write the full question number e.g. 2(B)(i) before the answer paragraph]

Answer any 5 of the following questions

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|-------|---|---|
| ✓ 1 A | What is serial and parallel communication? Explain with example. | 4 |
| 1 B | Define 8155 PPI. Write down the feature of 8155 PPI. | 5 |
| 1 C | How does inkjet printer work? Explain with figure. | 5 |
| ✓ 2 A | What is main difference between Sensors and Transforms? | 3 |
| 2 B | (i) Define Programmable Logic Controllers System. | 2 |
| | (ii) What is the purpose of opto isolators or optocoupler in a PLC? Explain with diagram of input /output channels. | 6 |
| 2 C | What do you mean by impact and non-impact printer ? | 3 |
| ✓ 3 A | What is an LVDT? How does LVDT work step by step? Explain with LVDT construction diagram. | 6 |
| 3 B | What is <u>asynchronous communication protocol</u> ? Briefly describe RS-232C connection with a <u>simple example</u> . | 8 |
| ✓ 4 A | Draw the functional block diagram of <u>8251A</u> PCI. | 3 |
| 4 B | Briefly explain functional block diagram of 8251A PCI with consisting sections. | 7 |
| 4 C | Which process of communication is controlled by an external circuit and also used for data transfer between memories and input/output ? Draw block diagram of a typical DMA controller. | 4 |
| 5 A | (i) What is the IEEE 488 bus system? | 2 |
| | (ii) Briefly explain the operation of IEEE 488 bus system based on the handshaking protocol with PIN configuration. | 6 |
| 5 B | How does a paper scanner work? Explain with functional diagram. | 6 |
| ✓ 6 A | <u>What is 82C55 PPI?</u> What will be the operation of stepper motor when it is connected with 82C55 PPI ? | 6 |
| 6 B | How does laser printer work? Explain with functional diagram. | 6 |
| 6 C | What is the example of biometric technology? | 2 |

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]
 Answer any 5 of the following questions:

1. (a). Define transducers. Write down the advantages of electrical signal as output of a transducer. 4
 (b). What do you mean by helipot? Describe the working procedure of L.V.D.T. 5
 (c). Describe the effect of variation of the dielectric constant for the measurement of displacement in a capacitive transducer. 5
2. (a). Define piezo-electric effect. Derive the equation of voltage sensitivity due to piezo-electric effect. 5
 (b). Write short notes on: i. Photoemissive cell ii. Photoconductive cell iii. Photovoltaic cell. 3
 (c). What is hall effect? Describe the working principle of hall effect transducers. 4
 (d). Write down the applications and limitations of thermistors. 2
3. (a). Define pyrometer. Describe the working principle of radiation pyrometer. Write down the advantages and limitations of radiation pyrometer. 6
 (b). What is thermocouple? Briefly discuss about the construction and operating principle of a thermocouple. 5
 (c). Write short notes on: i. U-type manometer ii. Well-type manometer. 3
4. (a). Describe Data Read and Data Write operations through FDC. 5
 (b). Encode data 101000100110 using FM, MFM and RLL encoding formats. Show the waveforms. 9
5. (a). Assume the following RAM chip and the microprocessor: 6



Connect the microprocessor to the RAM to obtain the memory map: 0000_{16} through $0FFF_{16}$. Use only the signals shown. Draw a neat logic diagram and analyze the memory map.

- (b). The MMU can perform address translation in one of the two ways. Define them. 3
- (c). Perform binary addition of $(-66)_{10}$ and $(-65)_{10}$. Now determine the carry, sign, overflow, and zero flags for this addition. 5
6. (a). What are the basic differences between ASCII and EBCDIC? 2
 (b). Explain the operation of a delta-delta shadow-mask CRT. 3
 (c). What do you mean by true color representation? Does it represent any disadvantages? 3
 (d). Write short notes on: 6
 - i. Interlaced refresh procedure
 - ii. 10H8
 - iii. Bitmap and Pixmap

(2016)

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Patuakhali Science and Technology University
6th Semester (L-3, S-II), Final Exam of B. Sc. Engg. (CSE), July -Dec. 2016
Course Code: CCE 321 Course Title: Computer Peripheral and Interfacing Session: 2013-2014
Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

2016

1. (a). Differentiate between MDA and CGA. 4
(b). Write down the advantages and disadvantages of LCD over CRT display. 4
(c). When a key on a keyboard is pressed, a switch is activated to generate signals for communicating the same to the PC. Give a brief description of four of them. 8
2. (a). What is key debouncing? Why is it needed? 2
(b). Define make code, break code, typematic, persistence, aspect ratio and resolution. 3
(c). Modern keyboards support bidirectional data transfer. How? 5
(d). Explain the phase sequences for the execution of "Test Unit Ready" SCSI command. 4
3. (a). Why is serial ATA used in present technologies instead of parallel ATA? 3
(b). Describe IDE command execution protocols. 10
(c). What do you mean by "CHRN" value? 1
4. (a). Describe Data Read and Data Write operations through FDC. 5
(b). Encode data 101000100110 using FM, MFM and RLL encoding formats. Show the waveforms. 9
5. (a). Illustrate an 8-bit microprocessor interfaced to a 4K RAM system using a full/partial decoded addressing technique. Also show the address map corresponding to the above mentioned organization. 5
(b). What do you mean by ISA bus and PCI bus? 2
(c). Write down the steps involved in starting a printing sequence using a printer controller chip. 4
(d). Explain the operation of a delta-delta shadow-mask CRT. 3
6. (a). What are the basic differences between ASCII and EBCDIC? 2
(b). Describe the process of a device initiated I/O transfer. 3
(c). When is an overflow flag set or reset? Give examples? 3
(d). Illustrate the character and signal generation procedure of a CRT display with a short description. 4
(e). The MMU can perform address translation in one of the two ways. Define them. 2

Memory

Patuakhali Science and Technology University

B.Sc.Engg.(CSE) L-3, S-2, Final Examination -2015 (July-Dec). Session: 2012-13

Course Code: CCE-321

Course Title: Computer Peripheral and Interfacing

Credit Hour: 3.0

Full Marks: 70

Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

[Use figures where necessary]

Answer any 5 of the following questions:

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1. (a). What is buffer? Why we use buffers? 3
(b). Define interfaces. What are the characteristics of computer peripherals? 4
(c). Differentiate between buffers and spoolers. 2
(d). What do you mean by solid state storage device? Write down the features and advantages of Solid state storage devices. 5
2. (a). Define raster scan display and random scan display. What are the differences between raster scan system and random scan system? 5
(b). What are the advantages of LCD panel over conventional CRT monitor? 4
(c). Differentiate between ATA RAID and SCSI RAID. 2
(d). Write down the selection criteria of a monitor. 3
3. (a). Define USB. What are the functions of USB ports? Write down the limitations of USB ports. 5
(b). Differentiate between IEEE1394 and USB. 3
(c). Sketch the three tier topology of USB. Write down the benefits of three tier topology of USB. 4
(d). What are the benefits of using USB? 2
4. (a). What is PCI Express? Write down the features and applications of PCI Express bus. 7
(b). Define resource conflict. How can user resolve resource conflict? 4
(c). Differentiate between serial ATA and parallel ATA. 3
5. (a). Differentiate between ISA, VESA and EISA buses. 5
(b). What do you mean by ATA? Write down the features of the ATA-1 specification. 4
(c). Write short note on: AGP, Plug and Play, SCSI. 3
(d). Differentiate between IRQ and DMA channels. 2
6. (a). Define RAID. Describe the different levels of RAID. 7
(b). Differentiate between IDE/ATA and SCSI. 4
(c). Write down the advantages of SCSI over ATA. 3

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) L-3, S-2, Final Examination -2014 (July-Dec), Session: 2011-12
Course Code: CCE-321 Course Title: Computer Peripheral and Interfacing

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

[Use figures where necessary]

Answer any 5 of the following questions:

1. (a). What is buffer? Why we use buffers? 4
(b). Define interfaces. What are the characteristics of computer peripherals? 4
(c). What do you mean by solid state storage device? Write down the features and advantages of Solid state storage devices. 6
2. (a). Define raster scan display and random scan display. What are the differences between raster scan system and random scan system? 6
(b). What are the advantages of LCD panel over conventional CRT monitor? 5
(c). Differentiate between IRQ and DMA channels. 3
3. (a). Define USB. Write down the limitations of USB ports. 5
(b). What are the benefits of using USB? 5
(c). Sketch the three tier topology of USB. Write down the benefits of three tier topology of USB. 4
4. (a). What is PCI Express? Write down the features and applications of PCI Express bus. 8
(b). Define resource conflict. How can user resolve resource conflict? 4
(c). What do you mean by voltage conversion and protocol conversion? 2
5. (a). Compare between VESA and PCI buses. 5
(b). What do you mean by ATA? Write down the features of the ATA-1 specification. 6
(c). Write short note on: AGP, Plug and Play, SCSI. 3
6. (a). Define RAID. Describe the different levels of RAID. 7
(b). Differentiate between IDE/ATA and SCSI. 4
(c). Write down the advantages of SCSI over ATA. 3

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

Dept. of Computer and Communication Engineering

Final Examination of B. Sc. Engineering In CSE Level: 3 Semester: II Session: 2018-2019

Course Code
CCE-322

Course Title
Computer Peripheral and Interfacing Sessional

July December 2021

Credit: 1.50
Marks: 70

1. Answer the following:

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- a. Illustrate the Gantt chart for your project.
- b. What is the function of the IOREF pin on the Arduino UNO?
- c. Arduino IDE consists of 2 functions. Define them.
- d. What is the use of the Arduino.h header file?
- e. What is the execution process of an Arduino code?
- f. Why does every compilation of a source code in Arduino check for previous compilations?
- g. What is the use of the ESP8266 WiFi Module?
- h. What is the way of throwing an error using preprocessing directives to the Arduino Compiler and forcing it to stop compilation?
- i. What will happen if we supply a voltage of 25V to the Vcc of the Nokia 5110 GDM?
- j. What is the output of "pin1" if "pin2" is sent "1011" where 1 is 5V and 0 is 0V?

```
1. int pin1 = 12;  
2. int pin2 = 11;  
3. void setup() {  
4.     pinMode(pin1, OUTPUT);  
5.     pinMode(pin2, INPUT);  
6.     Serial.begin(9600);  
7. }  
8. void loop() {  
9.     if(digitalRead(pin2)==1) {  
10.         digitalWrite(pin1,LOW);  
11.     }  
12.     else if(digitalRead(pin2)==0)  
13.         digitalWrite(pin1,HIGH);  
14.     }  
15. }
```

2. Make the project report.
3. Develop and display the Project.
4. Attend viva-voce.

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